MINISTRY OF EDUCATION AND RESEARCH
OF THE REPUBLIC OF MOLDOVA

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Report

Laboratory Work No. 2

Information Security and Cryptography

Cryptanalysis of Monoalphabetic Ciphers

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1 Objective

Study and practical application of frequency analysis techniques for breaking monoalphabetic substitution ciphers. Understand the vulnerabilities of monoalphabetic ciphers and develop skills in cryptanalysis.

2 Tasks

1. Study the concept of letter frequency analysis in different languages
2. Understand the methodology of frequency attack on monoalphabetic ciphers
3. Analyze and break a given cryptogram using frequency analysis
4. Recover the encryption key used in the cipher
5. Document the complete cryptanalysis process

3 Theoretical Background

3.1 Frequency Analysis Concept

The fundamental weakness of monoalphabetic encryption systems lies in the frequency of character appearance in the text. If an encrypted text is sufficiently long and the language of the plaintext is known, the system can be broken through a *frequency analysis attack*.

3.2 Letter Frequencies in Languages

English Language Frequencies:

The frequencies of the English language are:

E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

Figure 1: Execution times mean time

3.3 Frequency Attack Methodology

The most common digraphs in the english language are:

TH,HE,AN,IN,ER,ON,RE,ED,ND,HA,AT,EN

The most common digraphs in the message are:

WQ,QV,XG,VI,TG,VG,PW,QT,IV,NG,VP

Figure 2: The most common digraphs

The most common trigraphs in the english language are:

THE,AND,THA,ENT,ION,TIO,FOR,NDE,HAS,NCE,TIS,OFT,MEN

The most common trigraphs in the message are:

WQV,XGJ,TGO,VGW,QTO,XNG,RTP,KVI,QXP,TIO,WXN,PWT,TWX

Figure 3: The most common trigraphs

The most common double letters in the english language are:

SS,EE,TT,FF,LL,MM,OO

The most common double letters in the message are:

SS,VV,PP,NN,CC,WW,UU,GG,OO,HH

Figure 4: The most common double letters

4 Practical Cryptanalysis Example

4.1 Given Cryptogram [v21]

The intercepted cryptogram is known to be obtained using a monoalphabetic cipher over an English text:

```
Ftiosvf'p tuuinuixtwxng qto avvg pvkvivsf hdw xg 1924, tgo qtsc
    wqvpwtcc qto wn av svw jn, ivodhxgj wqv cnihv wn tandw t onmvg.
    Ovpuxwv wqxp,Ftiosvf ptxo, wqv Asthl Hqtzavi ztgtjvo wn pnskv, cinz
    1917 wn 1929,zniv wqtg 45,000 wvsvjitzp, xgknskxgj wqv hnovp nc
    Tijvgwxgt, Aitmxs,Hqxsv, Hqxgt, Hnpwt Ixht, Hdat, Vgjstgo, Citghv,
    Jviztgf, Etutg,Sxavixt, Zvyxhn, Gxhtitjdt, Utgtzt, Uvid, Ptg
    Ptsktoni, Ptgwn Onzxgjn(stwvi wqv Onzxgxhtg Ivudasxh) wqv Pnkxvw
    Dgxng, tgo Putxg tgo ztovuivsxzxgtif tgtsfpvp nc ztgf nwqvi hnovp,
    xghsdoxgj wqnpv nc wqv Ktwxhtg.Pdoovgsf xw tss vgovo. Ftiosvf, rqn
    qto avvg nawtxgxgj wqv hnovwvsvjitzp nc cnivxjg jnkvigzvgwp wqindjq
    wqv hnnuvitwxng nc wqvuivpxovgwp nc wqv Rvpwvig Dgxng Wvsvjituq
    Hnzutgf tgo wqv UnpwtsWvsvjituq Hnzutgf, rtp vghndgwvixgj xghivtpxgj
     ivpxpwtghv cinz wqvz.Qviaviw Qnnkvi qto edpw avvg xgtdjditwvo, tgo
    Ftiosvf ivpnskvo wn pvwwsv wqv ztwwvi rxwq wqv gvr tozxgxpwitwxng
    nghv tgo cni tss. Qvovhxovo ng wqv anso pwinlv nc oitrxgj du "t
    zvznitgodz wn avuivpvgwvo oxivhwsf wn wqv Uivpxovgw, ndwsxgxgj wqv
    qxpwnif tgo thwxkxwxvp ncwqv Asthl Hqtzavi, tgo wqv gvhvpptif pwvup
    wqtw zdpw av wtlvg xc wqvJnkvigzvgw qto qnuvo wn wtlv cdss tokvgwtjv
     nc wqv plxss nc xwphifuwnjituqvip." Qv rtxwvo wn pvv rqxhq rtf wqv
    rxgo rtp asnrxgj avcnivztlxgj qxp  z n k v tgo  cndgo wqtw xw rtp gnw
    rxwq qxz. Ftiosvf rvgw wn tpuvtlvtpf wn sxpwvg wn Qnnkvi'p cxipw
    puvvhq tp Uivpxovgw tgo pvgpvo, xgwqv qxjq vwqxhts pwixhwdivp wqtw
    Qnnkvi vyuivppvo, wqv onnz nc wqv AsthlHqtzavi.Qv rtp ixjqw, wqndjq
    xwp thwdts hsnpxgj htzv cinz vspvrqviv. TcwviQvgif S. Pwxzpng,
    Qnnkvi'p Pvhivwtif nc Pwtwv, qto avvg xg nccxhv wqv cvrzngwqp wqtw
    Ftiosvf wqndjqw rndso av gvhvpptif cni qxz wn qtkv snpwpnzv nc qxp
    xggnhvghv xg rivpwsxgj rxwq wqv qtioqvtovo ivtsxwxvp ncoxusnzthf,
    wqv Asthl Hqtzavi pvgw qxz wqv pnsdwxng nc tg xzuniwtgwpvixvp nc
    zvpptjvp. Adw Pwxzpng rtp oxccvivgw cinz uivkxndp Pvhivwtixvpnc
    Pwtwv, ng rqnz wqxp wthwxh qto tsrtfp rnilvo. Qv rtp pqnhlvo wnsvtig
     nc wqv vyxpwvghv nc wqv Asthl Hqtzavi, tgo wnwtssf oxptuuinkvo nc
    xw.Qv ivjtiovo xw tp t snr, pgnnuxgj thwxkxwf, t pgvtlxgj, pufxgj,
    lvfqnsv-uvvixgj lxgo nc oxiwf adpxgvpp, t kxnstwxng nc wqv uixghxusv
     nc zdwdts widpwdung rqxhq qv hngodhwvo anwq qxp uvipngts tcctxip
    tgo qxp cnivxjgunsxhf. Tss nc wqxp xw xp, tgo Pwxzpng ivevhwvo wqv
    kxvr wqtw pdhq zvtgpedpwxcxvo vkvg utwixnwxh vgop. Qv qvso wn wqv
    hngkxhwxng wqtw qxp hndgwifpqndso on rqtw xp ixjqw, tgo, tp qv ptxo
    stwvi, "Jvgwsvzvg on gnw ivtovthq nwqvi'p ztxs." Xg tg thw nc udiv
    znits hnditjv, Pwxzpng, tccxizxgjuixghxusv nkvi vyuvoxvghf, rxwqoivr
     tss Pwtwv Ovutiwzvgw cdgop cinz wqvpduuniw nc wqv Asthl Hqtzavi.*
    Pxghv wqvpv hngpwxwdwvo xwp ztenixghnzv, wqvxi snpp pqdwwvivo wqv
    nccxhv. Qnnkvi'p puvvhq qto rtigvoFtiosvf wqtw tg tuuvts rndso av
    cidxwsvpp. Wqviv rtp gnwqxgj wn on adwhsnpv du pqnu.Xg 1940, tp
    Pvhivwtif nc Rti, qv qto wn ivkvipv qxzpvsc tgo thhvuwwqv
    hifuwtgtsfpvp nc ZTJXH. Adw wqv xgwvigtwxngts pxwdtwxng wqvg
    rtpwnwtssf oxccvivgw. "Xg 1929," qv qxzpvsc qtp rixwwvg, xg wqv
    wqxio uvipng,"wqv rniso rtp pwixkxgj rxwq jnno rxss cni stpwxgj
    uvthv, tgo xg wqxp vccniwtss wqv gtwxngp rviv utiwxvp. Pwxzpng, tp
    Pvhivwtif nc Pwtwv, rtp ovtsxgjtp t jvgwsvztg rxwq wqv jvgwsvzvg
    pvgw tp tzatpptonip tgo zxgxpwvipcinz cixvgosf gtwxngp. ..." Xg
    1940, Vdinuv rtp tw rti, tgo wqv DgxwvoPwtwvp rtp ng wqv kvijv.Wqv
```

```
Pxjgts Hniup, rqviv Rxssxtz Cixvoztg qto hqtijv nc hifuwnsnjf.Wqv
pwtcc bdxhlsf oxpuvipvo (gngv rvgw wn wqv Tizf), tgo rqvg wqv
annlprviv hsnpvo ng Nhwnavi 31, 1929, wqv Tzvixhtg Asthl Hqtzavi
qtouvixpqvo. Xw qto hnpw wqv Pwtwv Ovutiwzvgw $230,404 tgo wqv
RtiOvutiwzvgw $98,808.49  edpw   dgovi t wqxio nc t zxssxng onsstip
cni tovhtov nc hifuwtgtsfxp. Ftiosvf, rqnpv ena vyuvixvghv qto avvg
itwqvi puvhxtsxmvo, hndso gnwcxgo rnil, tgo qv rvgw athl qnzv wn
Rniwqxgjwng. Wqv Ovuivppxngpdhlvo qxz oif. Af Tdjdpw nc 1930, qv qto
 qto wn jxkv du tg tutiwzvgwqndpv tgo t ngv-vxjqwq xgwvivpw xg t
ivws vpwtwv hniunitwxng; xgovvo, qvhnzustxgvo wqtw qv qto wn pvss
gvtisf vkvifwqxgj qv nrgvo "cni svpp wqtg gnwqxgj." T cvr zngwqp
stwvi qv rtp wnfxgj rxwq wqv xovt nc rixwxgj wqvpwnif nc wqv Asthl
Hqtzavi wn ztlv pnzv zngvf wn cvvo qxp rxcv tgowqvxi png, Ethl. Rqvg
 qxp nso ZX-8 cixvgo, Ztgsf, rxwq rqnz qv qtoavvg xg hngwthw tss
odixgj wqv 1920'p, qto wn wdig onrg qxp ivbdvpw cni t$2,500 sntg tw
wqv vgo nc Etgdtif, 1931, Ftiosvf, xg ovpuvitwxng, ptwonrg wn rixwv
rqtw rtp wn av wqv znpw ctzndp annl ng hifuwnsnjf vkviudasxpqvo.
```

4.2 Step 1: Frequency Analysis

First, we calculate the frequency of all letters in the cryptogram:

The frequencies of the English language are:

E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

The frequencies of the intercept are:

V	W	T	N	G	X	P	Q	I	O	S	H	C	Z	U	R	D	F	J	A	K	L	E	Y	M	B
439	334	309	262	252	234	224	218	210	164	134	115	91	91	77	72	69	68	67	51	33	27	9	5	3	2
12.3	9.4	8.7	7.4	7.1	6.6	6.3	6.1	5.9	4.6	3.8	3.2	2.6	2.6	2.2	2.0	1.9	1.9	1.9	1.4	0.9	0.8	0.3	0.1	0.1	0.1

Figure 5: Letter frequencies in the cryptogram

4.3 Step 3: Progressive Decryption

Figure 6: Progressive Decryption

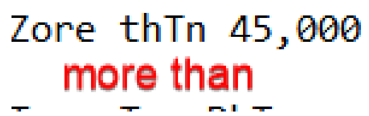

Figure 7: Progressive Decryption

Figure 8: Progressive Decryption

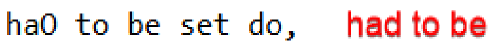

Figure 9: Progressive Decryption

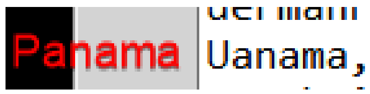

Figure 10: Progressive Decryption

Figure 11: Progressive Decryption

We gradually recover the complete substitution key.

4.4 Step 4: Final Decrypted Text

The complete decrypted message reveals

```
Yardley's appropriation had been severely cut in 1924, and half the
    staff had to be let go, reducing the force to about a dozen. Despite
     this, Yardley said, the Black Chamber managed to solve, from 1917
    to 1929, more than 45,000 telegrams, involving the codes of
    Argentina, Brazil, Chile, China, Costa Rica, Cuba, England, France,
    Germany, Japan, Liberia, Mexico, Nicaragua, Panama, Peru, San
    Salvador, Santo Domingo (later the Dominican Republic), the Soviet
    Union, and Spain and made preliminary analyses of many other codes,
    including those of the Vatican.

Suddenly it all ended. Yardley, who had been obtaining the code
    telegrams of foreign governments through the cooperation of the
    presidents of the Western Union Telegraph Company and the Postal
    Telegraph Company, was encountering increasing resistance from them.
     Herbert Hoover had just been inaugurated, and Yardley resolved to
    settle the matter with the new administration once and for all. He
    decided on the bold stroke of drawing up "a memorandum to be
    presented directly to the President, outlining the history and
    activities of the Black Chamber, and the necessary steps that must
    be taken if the Government had hoped to take full advantage of the
    skill of its cryptographers." He waited to see which way the wind
    was blowing before making his m o v e and found that it was not with
    him. Yardley went to a speakeasy to listen to Hoover's first speech
    as President and sensed, in the high ethical strictures that Hoover
    expressed, the doom of the Black Chamber.

He was right, though its actual closing came from elsewhere. After
    Henry L. Stimson, Hoover's Secretary of State, had been in office
    the few months that Yardley thought would be necessary for him to
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have lost some of his innocence in wrestling with the hardheaded realities of diplomacy, the Black Chamber sent him the solution of an important series of messages. But Stimson was different from previous Secretaries of State, on whom this tactic had always worked. He was shocked to learn of the existence of the Black Chamber, and totally disapproved of it. He regarded it as a low, snooping activity, a sneaking, spying, keyhole-peering kind of dirty business, a violation of the principle of mutual trust upon which he conducted both his personal affairs and his foreign policy. All of this it is, and Stimson rejected the view that such means justified even patriotic ends. He held to the conviction that his country should do what is right, and, as he said later, "Gentlemen do not read each other's mail." In an act of pure moral courage, Stimson, affirming principle over expediency, withdrew all State Department funds from the support of the Black Chamber.* Since these constituted its major income, their loss shuttered the office. Hoover's speech had warned Yardley that an appeal would be fruitless. There was nothing to do but close up shop.

In 1940, as Secretary of War, he had to reverse himself and accept the cryptanalyses of Magic. But the international situation then was totally different. "In 1929," he himself has written, in the third person, "the world was striving with good will for lasting peace, and in this effort all the nations were parties. Stimson, as Secretary of State, was dealing as a gentleman with the gentlemen sent as ambassadors and ministers from friendly nations. ..." In 1940, Europe was at war, and the United States was on the verge.

The Signal Corps, where William Friedman had charge of cryptology. The staff quickly dispersed (none went to the Army), and when the books were closed on October 31, 1929, the American Black Chamber had perished. It had cost the State Department $230,404 and the War Department $98,808.49 just under a third of a million dollars for a decade of cryptanalysis. Yardley, whose job experience had been rather specialized, could not find work, and he went back home to Worthington. The Depression sucked him dry. By August of 1930, he had had to give up an apartment house and a one-eighth interest in a real estate corporation; indeed, he complained that he had to sell nearly everything he owned "for less than nothing." A few months later he was toying with the idea of writing the story of the Black Chamber to make some money to feed his wife and their son, Jack. When his old MI-8 friend, Manly, with whom he had been in contact all during the 1920's, had to turn down his request for a $2,500 loan at the end of January, 1931, Yardley, in desperation, sat down to write what was to be the most famous book on cryptology ever published.

4.5 Step 5: Recovered Encryption Key

V	W	T	N	G	X	P	Q	I	O	S	H	C	Z	U	R	D	F	J	A	K	L	E	Y	M	B
439	334	309	262	252	234	224	218	210	164	134	115	91	91	77	72	69	68	67	51	33	27	9	5	3	2
12.3	9.4	8.7	7.4	7.1	6.6	6.3	6.1	5.9	4.6	3.8	3.2	2.6	2.6	2.2	2.0	1.9	1.9	1.9	1.4	0.9	0.8	0.3	0.1	0.1	0.1
e	t	a	o	n	i	s	h	r	d	l	c	f	m	p	w	u	y	g	b	v	k	j	x	z	q

Figure 12: Decrypted alphabet

5 Results and Analysis

5.1 Cryptanalysis Process Documentation

The cryptanalysis followed this systematic approach:

1. **Frequency Counting:** Calculated exact frequencies of all ciphertext letters
2. **Initial Mapping:** Matched most frequent ciphertext letters with most frequent English letters
3. **Pattern Recognition:** Identified common words and patterns (”THE”, ”TO”, ”ONE”)
4. **Contextual Analysis:** Used word context to resolve ambiguities
5. **Iterative Refinement:** Progressively refined the substitution key
6. **Validation:** Verified the decrypted text made linguistic sense

5.2 Key Recovery

The recovered substitution key demonstrates the complete breakdown of the monoalphabetic cipher. The mapping shows a non-standard substitution pattern rather than a simple Caesar shift.

6 Conclusions

6.1 Main Findings

1. Monoalphabetic ciphers are highly vulnerable to frequency analysis attacks
2. The effectiveness increases with longer ciphertexts due to statistical convergence
3. Additional linguistic patterns (digraphs, trigraphs, common words) are crucial for successful cryptanalysis
4. Human judgment remains essential in the cryptanalysis process

6.2 Security Implications

- Monoalphabetic ciphers provide no real security for sensitive communications
- Even with large keyspaces (26! possible keys), the statistical properties of language make them vulnerable
- These ciphers are only suitable for educational purposes or very low-security applications

6.3 Practical Applications

- Understanding fundamental cryptanalysis techniques
- Demonstrating the importance of eliminating statistical patterns in secure encryption
- Foundation for understanding more advanced cryptographic attacks
- Historical perspective on the evolution of cryptography

This laboratory work successfully demonstrated that monoalphabetic substitution ciphers, while having a large theoretical keyspace, can be efficiently broken through frequency analysis due to the inherent statistical properties of natural languages.

Source Code

https://github.com/Nickseen/Cryptology-Security/tree/Lab2